

Science in the Skies: Hot Air Balloons Take Center Stage in New Clark City

NEW CLARK CITY, TARLAC — Skies across New Clark City, Tarlac filled with vibrant colors as the Philippine International Hot Air Balloon Fiesta (PIHABF) welcomed aviation enthusiasts, students, and tourists to the province. Beyond offering a stunning visual spectacle, the event serves as an engaging outdoor laboratory that showcases fluid dynamics and thermodynamics—specifically Archimedes' Principle and thermal expansion.

The main attraction—massive hot air balloons rising effortlessly into the air—operates directly on Archimedes' Principle, which states that an object immersed in a fluid (or air) is buoyed upward by a force equal to the weight of the fluid it displaces. Ground crews use powerful burners to heat the air inside the balloon's envelope. As the air heats up, its molecules expand and spread apart, making the air inside less dense than the cooler ambient air outside. This difference in density creates a net upward buoyant force, causing the balloon to lift off the ground.

Aerodynamics and fluid mechanics also govern the flight path once the balloons are airborne. Pilots manage ascent and descent by controlling the temperature of the internal air, while navigating horizontal directions by tapping into varying wind currents at different altitudes.

Events at the venue—including flying drone demonstrations, ultra-light aircraft flights, and kite-flying competitions—further demonstrate forces such as lift, drag, thrust, and gravity working in harmony.

The New Clark City aviation showcase proves that physics isn't confined to a classroom. Behind every floating balloon and soaring aircraft are fundamental scientific laws powering the excitement high above Tarlac.